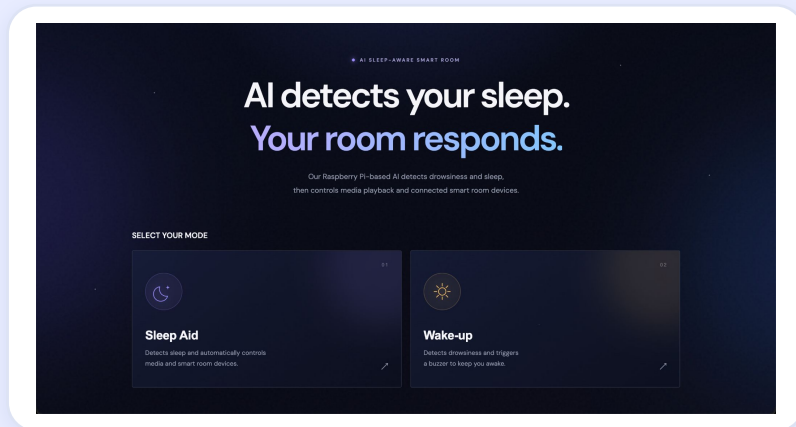




AI-Based Sleep-Aware Media and Smart Room Control System

Drowsiness Detection · Smart Room Automation · Web-Based Mode Control





Motivation: Why Sleep-Aware Automation?

01 Watching Video Before Sleep

Problem:

Video & lamp remain ON after the user falls asleep

Auto: Lamp OFF + Media Pause

02 Studying / Working While Sleepy

Problem:

No alert system to keep the user awake

Auto: Buzzer Alert + Media Pause

Key Insight: The same drowsy state requires different responses based on the user's intention. **Sleep Aid Mode** helps the user sleep comfortably. **Wake-up Mode** keeps the user alert.



Project Objectives

Build an AI-based smart room system that detects eye closure and controls connected devices automatically.

01 Web Interface

Browser-based UI for mode and email configuration (auto-starts camera)

02 Detection & Classification

Detect face & eyes using camera and classify AWAKE, EYES CLOSED, DROWSY

03 Lamp Control

Control USB mood lamp via relay module

04 Media Control

Auto pause/resume YouTube via AppleScript

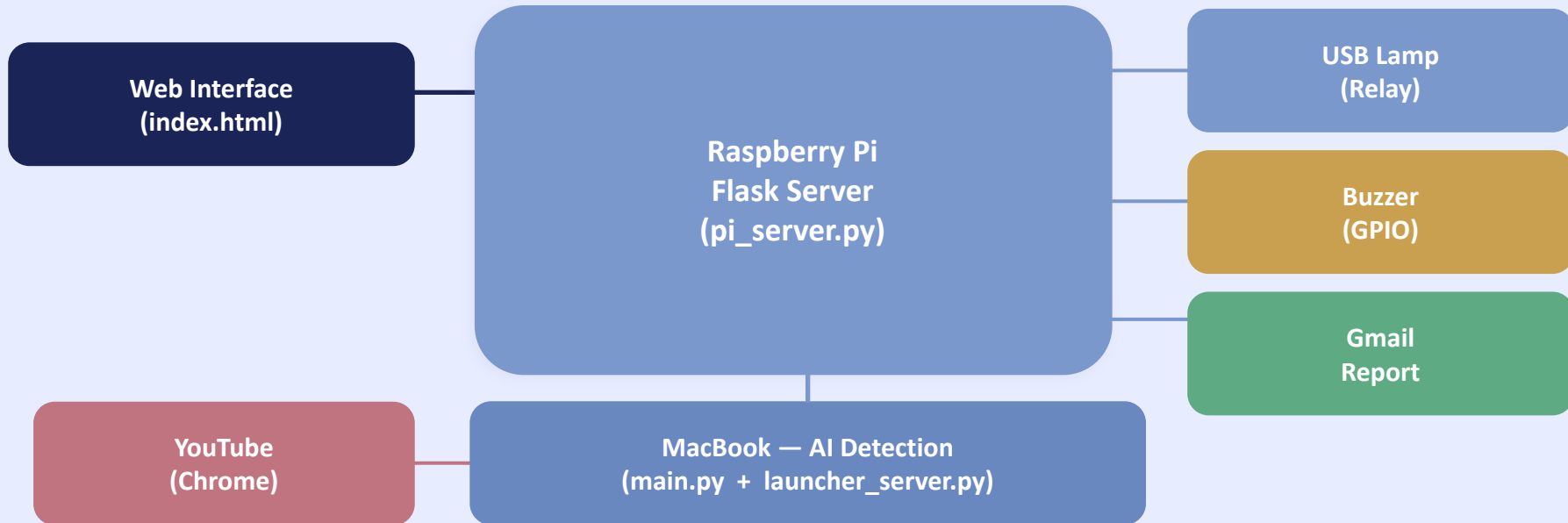
05 Buzzer Alert

Sound buzzer in Wake-up Mode

06 Email Report

Send sleep report via Gmail SMTP

System Overview



Note: MediaPipe could not be installed on Raspberry Pi, so AI detection runs on MacBook.

Raspberry Pi handles Flask API, GPIO control (lamp/buzzer), and email.

MacBook handles camera detection and YouTube control via AppleScript.



Hardware & Software Stack

Hardware Components

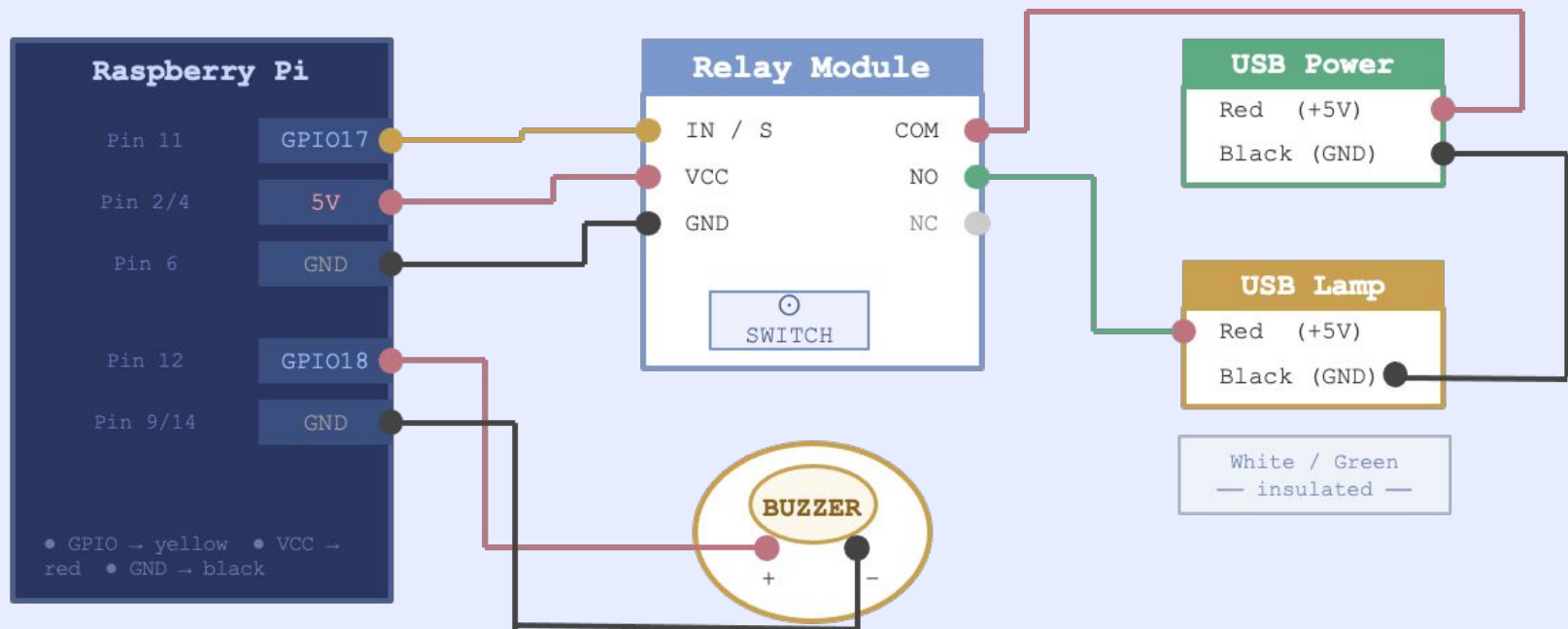
Raspberry Pi	IoT control server
Camera	Face & eye detection input
Relay Module	USB lamp power control
USB Mood Lamp	Smart room lighting output
Buzzer	Wake-up alert output
USB Extension Cable	Relay-based lamp power control

Tip: +5V line of USB extension cut → relay COM/NO → controls lamp power.A

Software Stack

AI Detection	OpenCV · MediaPipe Face Mesh · NumPy
IoT Control	Flask · RPi.GPIO · Relay / Buzzer
Web Interface	HTML · CSS · JavaScript
Media Control	AppleScript → Chrome YouTube
Notification	Gmail SMTP (Sleep Report Email)
Launcher	Flask launcher_server.py

Raspberry Pi Circuit Diagram



Wire Legend

- | | |
|--------------------------|--------------------------------------|
| GPIO Signal (Yellow) | Relay NO → Lamp switched +5V (Green) |
| +5V Power / GPIO18 (Red) | Insulated / Not Used (White, Green) |
| GND / Ground (Black) | |

How Lamp Control Works
GPIO17 → Relay coil energizes
COM-NO closes → +5V flows → Lamp ON



Drowsiness Detection Using Eye Aspect Ratio (EAR)

EAR Formula

$$\text{EAR} = \frac{(\|p2-p6\| + \|p3-p5\|)}{(2 \times \|p1-p4\|)}$$

Ratio of eye height to width
to determine eye open/closed state.

Key Logic (main.py):

```
if closed_frames >= DROWSY_STATUS_THRESHOLD: # 60 frames
    status = "DROWSY"
elif closed_frames > 0:
    status = "EYES CLOSED"
else:
    status = "AWAKE"
```

AWAKE

EAR $\geq$ 0.21
Frames = 0
No action

EYES CLOSED

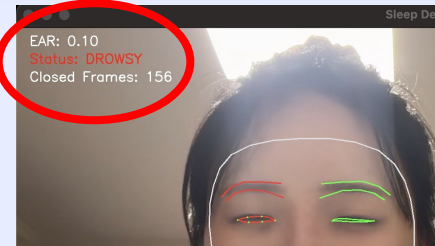
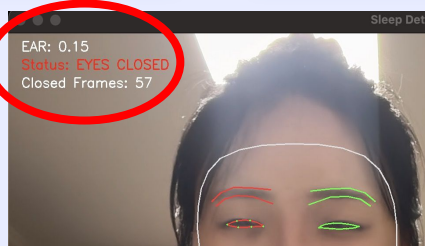
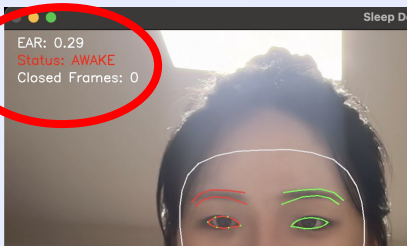
EAR < 0.21
Frames 1–59
Blink only

DROWSY

Frames $\geq$ 60
Status shown
Warning

ACTION!

Frames $\geq$ 61
Mode-specific
action runs





Two Operation Modes

Sleep Aid Mode

For watching videos before sleep

- DROWSY → Lamp turns OFF
- DROWSY → Media pauses
- DROWSY → No buzzer
- AWAKE → Sleep report email sent

DROWSY: start sleep timer | AWAKE: end timer + send email

Wake-up Mode

For studying or working

- DROWSY → Buzzer alert sounds
- DROWSY → Lamp stays ON
- DROWSY → Media pauses
- AWAKE → Media resumes

DROWSY: buzzer + media off | AWAKE: resume media

Same DROWSY state — different responses based on the user's intention.



Web-Based Mode and Email Configuration

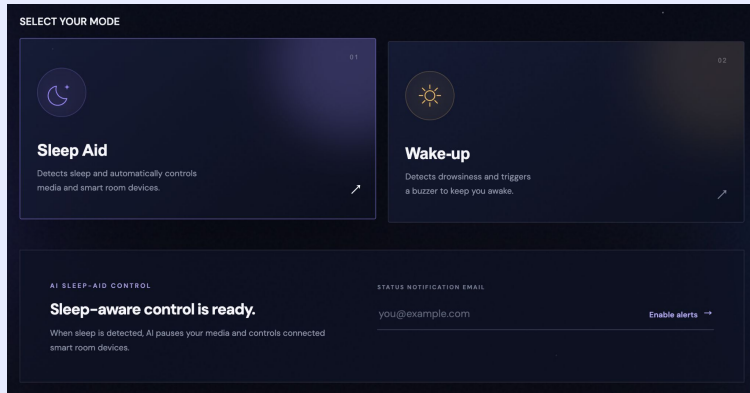
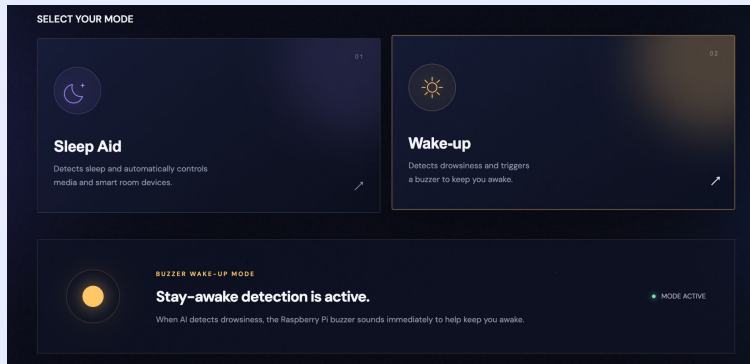
Web UI Features

- Sleep Aid Mode button / Wake-up Mode button
- Email input field for sleep reports
- JavaScript API call to save mode & email
- Auto-launch camera

Key Code — script.js:

```
// Sleep Aid Mode
await sendConfig("sleep", email);
await startDetection();

// Wake-up Mode
await sendConfig("wake");
await startDetection();
```



Improvement: terminal mode input → browser-based UI with mode buttons + email form.



Communication Flow & API Endpoints

Communication Flow

- 1 User selects a mode on the Web UI
- 2 The selected mode and email are saved to the Pi server
- 3 The detection program is launched on the Host PC
- 4 The AI detection system loads the selected mode
- 5 The camera detects the user's eye state
- 6 DROWSY triggers a mode-specific request
- 7 Raspberry Pi controls lamp, buzzer, or email

API Endpoints (pi_server.py)

<code>/set_config</code>	Save mode & email
<code>/get_config</code>	Provide mode to detection program
<code>/reset</code>	Lamp ON, buzzer OFF
<code>/sleep_start</code>	Start Sleep Aid action
<code>/sleep_end</code>	End + send email report
<code>/wake_drowsy</code>	Wake-up alert action
<code>/light_on /light_off</code>	Lamp control
<code>/buzzer</code>	Buzzer test



Key Implementation Details

1 Mode-Based Action Control (main.py)

```
if MODE == "sleep":
    send_request("sleep_start")
    pause_youtube()
    sleep_session_active = True
elif MODE == "wake":
    send_request("wake_drowsy")
    pause_youtube()
```

2 Sleep Report Trigger (main.py)

```
# When user wakes up after sleep session
if MODE == "sleep" and sleep_session_active:
    send_request("sleep_end")
    sleep_report_sent = True
    sleep_session_active = False
```

3 Gmail SMTP Email Report (pi_server.py)

```
with smtplib.SMTP_SSL("smtp.gmail.com", 465) as server:
    server.login(EMAIL_ADDRESS, EMAIL_PASSWORD)
    server.send_message(msg)
# msg includes: sleep start time, wake-up time, duration, activated functions
```

main.py AI detection + YouTube media control

script.js Web interface → backend API bridge

pi_server.py Flask API + GPIO control + email report

launcher_server.py Starts main.py when web UI activates



Demonstration Scenario 1: Sleep Aid Mode

Steps

- 1 Select Sleep Aid Mode on web page
- 2 Enter the report email address
- 3 Camera starts automatically
- 4 User closes eyes
- 5 DROWSY detected (frames ≥ 61)
- 6 Lamp turns OFF, YouTube pauses
- 7 User opens eyes — AWAKE
- 8 Sleep report email is sent

Result: Lamp OFF · Media paused · Email sent

AI SLEEP-AID CONTROL

Sleep-aware control is ready.
When sleep is detected, AI pauses your media and controls connected smart room devices.

STATUS NOTIFICATION EMAIL

lleeeynn137@gmail.com
Sleep-Aid status alerts will be sent to lleeeynn137@gmail.com.

Enable alerts →

Sleep Data

EAR: 0.10
Status: DROWSY
Closed Frames: 156

[Sleep-Aware System] Sleep Report Inbox x

2026-06-20, 10:36 PM (1 day ago)

dlgu5tn4665@gmail.com
to me ▾

Sleep-Aware System Report

Mode: Sleep-Aid Mode

Sleep detected time:
2026-06-20 22:36:35

Wake-up detected time:
2026-06-20 22:36:49

Estimated sleep duration:
13.9 seconds
(0.23 minutes)

Activated functions:
- Lamp turned off
- Media playback paused
- Buzzer was not activated

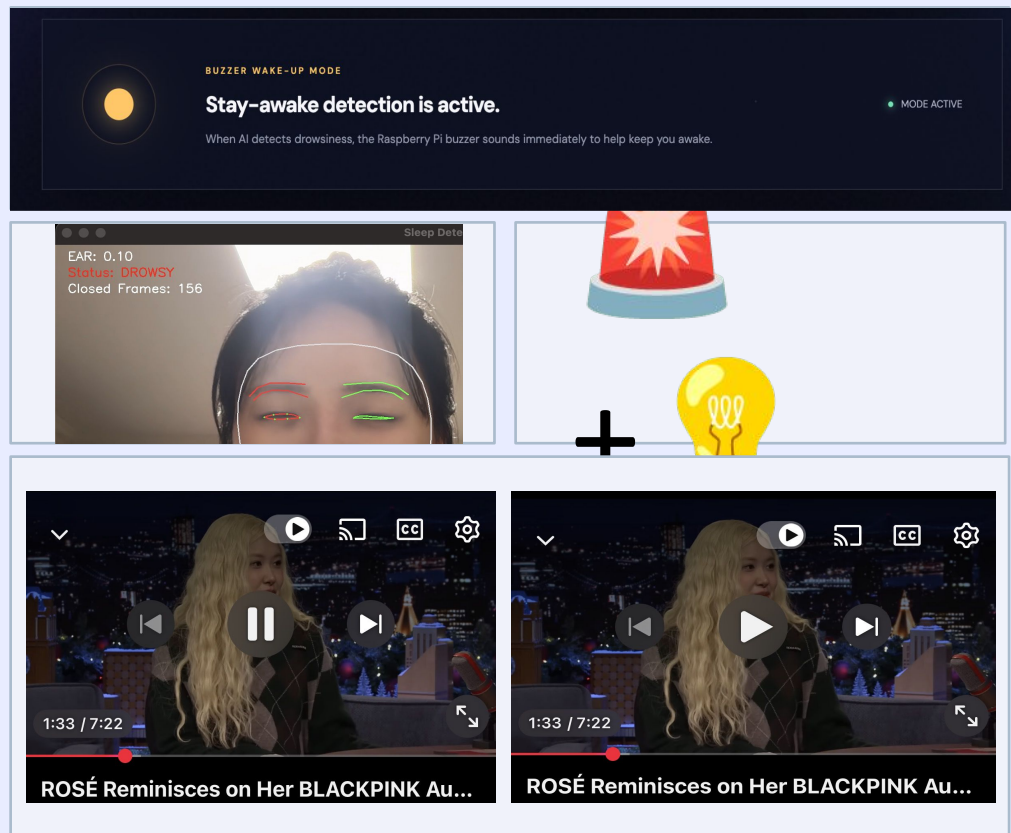


Demonstration Scenario 2: Wake-up Mode

Steps

- 1 Select Wake-up Mode on web page
- 2 Camera starts automatically
- 3 User closes eyes
- 4 DROWSY detected (frames ≥ 61)
- 5 Buzzer alert is activated
- 6 Lamp stays ON
- 7 YouTube pauses
- 8 User opens eyes — YouTube resumes

Result: Buzzer ON · Lamp ON · Media resumes on wake





Results & Challenges

Completed — Face & Eye Detection

Completed — State Classification

Completed — Sleep Aid Mode

Completed — Wake-up Mode

Completed — Lamp Control

Completed — Buzzer Control

Completed — YouTube Control

Completed — Email Report

Completed — Web UI + Auto Launch

Key Challenges & Solutions

MediaPipe installation issue on Raspberry Pi

→ Moved AI to MacBook; Pi handles IoT control only

Relay trigger instability with Raspberry Pi GPIO

→ Used active-low open-drain style control

USB lamp power control

→ Cut +5V line, connected through relay COM/NO

YouTube keyboard control unstable

→ Used AppleScript to control Chrome video element directly

Web cannot run Python directly

→ Added launcher_server.py as Flask bridge

Email sending timeout

→ Increased timeout; used Gmail SMTP app password



Limitations & Conclusion

Limitations

- Detection depends on lighting & camera angle
- YouTube control depends on Chrome and macOS
- Gmail SMTP needs app password setup

Future Work

- Improve detection robustness under various lighting and viewing angles
- Develop a cross-platform media control for Windows and other devices
- Mobile / KakaoTalk / Discord notifications
- Integrate with commercial smart home devices
- Sleep pattern dashboard visualization

This project integrates AI-based drowsiness detection, Raspberry Pi hardware control, web-based configuration, media control, and email reporting. With Sleep Aid Mode and Wake-up Mode, the system responds differently based on the user's intention.

AI detects the user's state. The room responds automatically.

Thank you